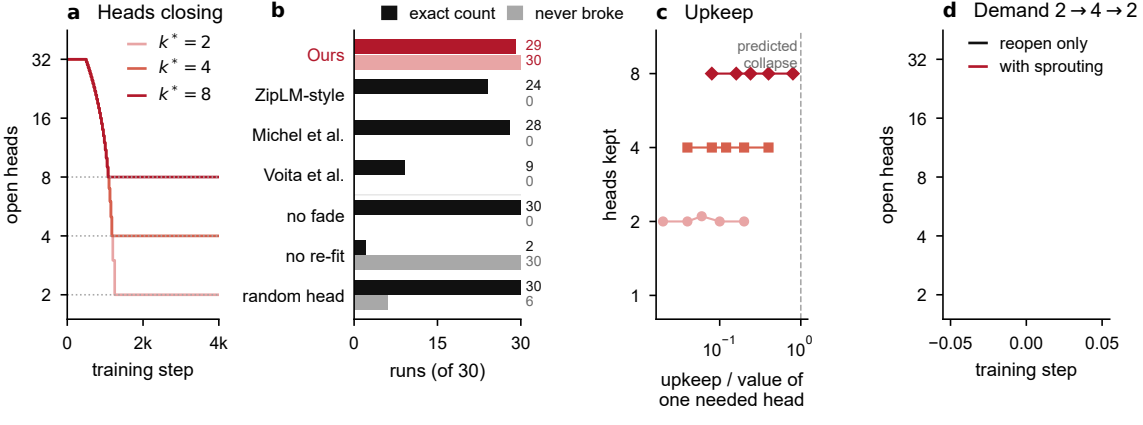
a
Heads closing
$k^* = 2$
$k^* = 4$
$k^* = 8$
open heads
32
16
8
4
2
0
2k
4k
training step
b
exact count
never broke
Ours
ZipLM-style
Michel et al.
Voita et al.
no fade
no re-fit
random head
29
30
24
0
28
0
9
0
30
0
2
30
30
6
0
15
30
runs (of 30)
c
Upkeep
predicted
collapse
heads kept
8
4
2
1
10^{-1}
10^{0}
upkeep / value of
one needed head
d
Demand 2 → 4 → 2
reopen only
with sprouting
open heads
32
16
8
4
2
−0.05
0.00
0.05
training step